Professional Ethics & Project Management

1. **Explain the concepts of morals, values, and ethics. How do they differ and intersect?**

Ans:

Morals:

It is a personal belief about what is right and wrong. It comes from family, culture, religion, etc.

Value:

Values are things that matter in life.

For example: we value honesty, respect, and success.

Ethics:

It is a formal rules or guidelines about what is right and wrong.

For example: A doctor must follow medical ethics and treat the patient.

How they are connected:

Values:

What we care about the most

Ex: We value honesty

Morals:

What we believe is right or wrong

Ex: Because we value honesty, we believe lying is wrong

Ethics:

Rules everyone follows

Ex: A teacher should treat all students equally

Moral	Value	Ethics
It is a personal belief about what is right and wrong. It comes from family, culture, religion, etc.	Values are things that matter in life.	It is a formal rules or guidelines about what is right and wrong.
Source – Personal belief	Source- Personal priorities	Source- Society
Type- Internal	Type- Internal	Type- External
Eg: Stealing is wrong	Eg: Honesty is important	Eg: Don't cheat in exam

2. **Discuss the significance or Importance of engineering ethics and professionalism in the field of engineering.**

Ans:

<u>Significance/ Importance of Engineering Ethics and Professionalism:</u>

i. **Keeps people safe:** Engineers make bridges, machines, buildings—so they must avoid mistakes.
ii. **Builds trust:** People trust engineers who are honest and do the right thing.
iii. **Makes engineers responsible:** They take responsibility for their work and decisions.
iv. **Improves work quality:** Being professional means doing work carefully and correctly.
v. **Stops wrong practices:** Ethics helps avoid cheating, corruption, and misuse of power.
vi. **Protects environment:** Engineers think about nature and future generations.
vii. **Good reputation:** Ethical engineers are respected in society.

3. **Describe the importance of a code of ethics in professional practice. Provide examples of codes of ethics from different professions.**

Ans:

<u>Importance of Code of Ethics in Professional Practice:</u>

A code of ethics is a set of rules that tells professionals how to behave properly.

i. **Gives clear guidelines:** Helps people know what is right and wrong in their job.
ii. **Builds trust:** People trust engineers, doctors, teacher, lawyer, businessman who are honest and do the right thing.
iii. **Ensures responsibility:** Professionals take responsibility for their work and decisions.
iv. **Improves work quality:** Being professional means doing work carefully and correctly.
v. **Stops wrong practices:** Ethics helps avoid cheating, corruption, and misuse of power.
vi. **Protects people and society:** Ensures safety, privacy, and well-being.

<u>Examples of Codes of Ethics:</u>

i. **Engineering:** Engineers must ensure safety, avoid harm, and be honest in their work.
ii. **Medical (Doctors):** Doctors must treat patients equally and save lives.
iii. **Business:** Companies must be fair, avoid fraud, and follow laws.
iv. **Teaching:** Teachers should treat all students equally and help them learn honestly.
v. **Law (Lawyers):** Lawyers must maintain client confidentiality and follow justice.

4. **Compare and contrast different models of professional roles in engineering.**

Ans:

Models of Engineers:

Model	What it means	Main focus
Savior	Engineer solves problems like a hero	New ideas, innovations
Gurdian	Engineer protects people and environment	Safety
Bureaucratic	Engineer follows rules and procedures	Rules, regulations
Social Servant	Engineer works for society benefit	People's welfare

5. **Explain the concept of whistleblowing and its ethical implications in the context of engineering.**

Ans:

Whistleblowing means reporting wrong or illegal activities happening in an organization. It is usually done by an employee to protect people and society.

Example:

An employee finds that a company is using unsafe materials in construction. He reports it to higher authority or government. This is whistleblowing

Types of Whistleblowers:

i. **Internal Whistleblower**

Reports the problem inside the organization

Example: Telling the manager or HR

ii. **External Whistleblower**

Reports the problem outside the organization

Example: Going to media or government

iii. **Open Whistleblower**

Reveals their identity openly

iv. **Anonymous Whistleblower**

Hides their identity safety

6. Analyse global ethical issues faced by multinational corporations, with a focus on environmental ethics and business ethics.

Ans:

<u>Global Ethical Issues in Multinational Corporations (MNCs):</u>

MNCs are companies that work in many countries. They face ethical problems because rules and conditions differ from place to place.

Environmental Ethics:

i. Some companies pollute air, water, and land in developing countries.
ii. Excessive use of water, minerals, and forests.
iii. Dumping harmful waste without proper treatment.
iv. Increasing global warming due to industries.

Example: A factory releasing harmful chemicals into rivers.

Business Ethics:

i. Paying workers very little in poor countries.
ii. Unsafe workplaces and long working hours.
iii. Bribery to get contracts or avoid laws.
iv. Mislead customers with false ads.

Example: Using cheap labour without proper safety measures.

7. Explain the concept of moral leadership and its significance in engineering practice.

Ans:

Moral leadership means leading people in a right, honest, and fair way. A moral leader does the right thing, respects others, and sets a good example for everyone.

Significance in engineering practice:

i. **Keeps people safe:** Engineers make bridges, machines, buildings—so they must avoid mistakes.
ii. **Builds trust:** People trust engineers who are honest and do the right thing.
iii. **Makes engineers responsible:** They take responsibility for their work and decisions.
iv. **Improves work quality:** Being professional means doing work carefully and correctly.
v. **Stops wrong practices:** Ethics helps avoid cheating, corruption, and misuse of power.
vi. **Protects environment:** Engineers think about nature and future generations.
vii. **Good reputation:** Ethical engineers are respected in society.

8. **Define project management and discuss its importance in various industries. Identify and discuss the issues and problems commonly encountered in project management.**

Ans:

<u>Project Management:</u>

Project management means planning, organizing, and completing a task (project) on time, within budget, and with good quality.

<u>Example:</u> Building a bridge, developing software, or organizing an event.

<u>Importance in Different Industries:</u>

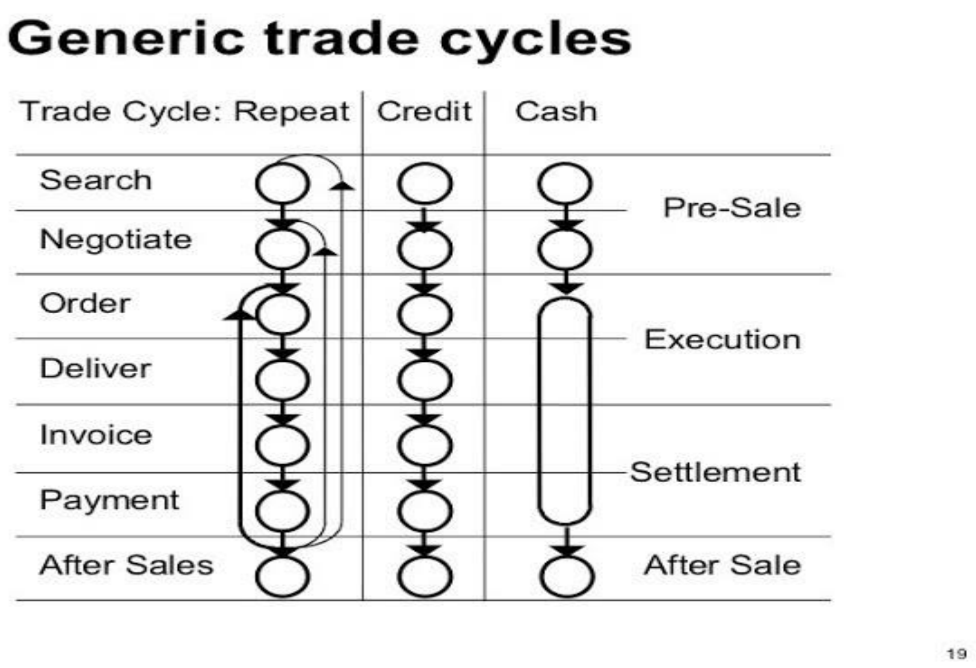
i. **Construction:** Helps complete buildings safely and on time.
ii. **IT/Software:** Ensures software is developed correctly and delivered on time.
iii. **Healthcare:** Helps manage hospital work properly.
iv. **Business:** Helps launch products and manage operations smoothly.

<u>Common Issues in Project Management:</u>

i. Project takes more time than planned.
ii. Costs become higher than expected.
iii. Work is not properly planned.
iv. Team members don't share information properly.
v. Not enough workers, tools, or materials.

9. **Describe the phases of the project life cycle and explain the activities involved in each phase.**

Ans:



Phases of Project Life Cycle:

i. **Initiation (Start):**
   a. Decide the project idea
   b. Check if it is possible
   c. Define goals

   Example: Deciding to build a bridge

ii. **Planning:**
   a. Make a plan of work
   b. Decide time, cost, and resources
   c. Assign tasks

   Example: Making design, budget, and schedule

iii. **Execution (Work starts):**
   a. Do the actual work
   b. Team members perform tasks

   Example: Construction of the bridge

iv. **Monitoring & Control:**
   a. Check progress of work
   b. Fix problems
   c. Ensure project is on time and budget

   Example: Checking if construction is going as planned

v. **Closure (Finish):**
   a. Complete the project
   b. Hand over the project

   Example: Opening the bridge for public use

10. Explain the concepts of Work Breakdown Structure (WBS) and Organization Breakdown Structure (OBS) in project planning.

Ans:

Work Breakdown Structure (WBS):

WBS means dividing a big project into small, simple tasks so it is easier to manage.

a. Breaks work step by step
b. Helps in planning and tracking
c. Makes work clear and organized

Example: Building a house → design → foundation → walls → roof → finishing

Organization Breakdown Structure (OBS):

OBS means dividing people or teams based on their roles and responsibilities in the project.

a. Shows who is responsible for what
b. Helps in proper management
c. Avoids confusion among team members

Example: Manager → Engineers → Workers → Support staff

11. Discuss the significance of project monitoring and control. Describe the role of a project manager and the use of Management Information Systems (MIS) in project monitoring.

Ans:

Project monitoring and control means checking the project regularly and fixing problems.

Significance of Project Monitoring and Control:

i. Ensures work is going as planned
ii. Saves time and cost
iii. Makes sure work is done correctly
iv. Helps complete project on time
v. Gives clear information about progress

Role of Project Manager:

i. Plans the project
ii. Assigns work to team members
iii. Monitors progress regularly
iv. Communicates with team and stakeholders
v. Ensures project success

Role of MIS (Management Information System):

MIS is a system that collects and shows project data.

i. Tracks progress of tasks
ii. Stores information
iii. Gives reports for decision-making
iv. Improves communication
v. Helps in quick problem solving

12. Discuss the importance of feasibility studies in project management.

Ans:

Feasibility Study:

Feasibility study means checking if a project is possible and worth doing before starting it. It looks at cost, time, resources, and risks to decide whether the project should go ahead or not.

Importance of Feasibility Study:

i. **Avoids loss:** Helps avoid wasting time and money on bad projects
ii. **Better decisions:** Shows whether the project should be done or not
iii. **Checks resources:** Ensures enough money, people, and materials are available
iv. **Reduces risk:** Identifies problems early
v. **Improves planning:** Helps make a clear and proper plan
vi. **Ensures success:** Increases chances of completing the project successfully

13. Discuss the ethical implications of cost-benefit analysis in project feasibility studies. Explain how economic considerations can influence project decisions.

Ans:

Cost-benefit analysis means comparing cost (money spent) and benefit (profit or gain) before starting a project.

Ethical Implications of Cost-Benefit Analysis:

a) Profit over people

Companies may focus more on earning money and ignore the needs of workers, customers, or society.

b) Environmental damage

To reduce costs, companies may pollute air, water, or land, harming nature.

c) Safety and health issues

To save money, companies may reduce safety measures, which can be dangerous.

d) Human values cannot be measured

Things like human life, health, and environment cannot be properly measured.

e) Unfair decisions

Benefits may go to a few people, while costs are faced by many others.

How Economic Considerations Influence Project Decisions:

a) **Preference for high-profit projects**

Projects that give more profit are selected over others.

b) **Cost reduction focus**

Companies try to reduce costs, sometimes affecting quality and safety.

c) **Budget limitations**

Projects are chosen based on available money and financial limits.

d) **Return on investment (ROI)**

Projects with quick and high returns are preferred.

e) **Rejection of low-profit projects**

Projects that are socially useful but less profitable may not be selected.

f) **Short-term thinking**

Companies may focus on short-term gains instead of long-term benefits.

14. Evaluate the ethical responsibilities of engineers in multinational corporations, considering factors such as labour practices, environmental impact, and community engagement.

Ans:

Ethical Responsibilities of Engineers in MNCs:

i) **Labor Practices:**

- Treat workers fairly
- Provide safe working conditions
- Give fair wages
- Avoid child labour

Example: Not forcing workers to work long hours in unsafe factories

ii) **Environmental Impact:**

- Reduce pollution
- Use resources carefully
- Follow environmental laws
- Support sustainable development

Example: Not dumping waste into rivers

iii) Community Engagement:

- Respect local culture and people
- Support local development (jobs, education)
- Avoid harming nearby communities

Example: Building projects without disturbing local residents

15. What is the role of a project manager in project monitoring and control?

Ans:

Role of a Project Manager in Monitoring and Control:

A project manager makes sure the project is going as planned and solves problems if needed.

i. **Tracks progress** – Checks if work is going according to plan
ii. **Controls time** – Makes sure the project finishes on time
iii. **Manages budget** – Keeps spending within limits
iv. **Maintains quality** – Ensures good quality work
v. **Identifies problems** – Finds issues early
vi. **Takes corrective action** – Fixes problems quickly
vii. **Reports status** – Updates stakeholders about progress

16. Explain the concept of project scope creep and its ethical implications.

Ans:

Project Scope Creep:

Project scope creep means adding extra work or changes to a project without proper approval or planning after it has already started.

Example: A client keeps asking for new features, but does not give extra time or money

Ethical Implications:

i. **Unfair workload –** Team has to do more work without extra time or pay
ii. **Quality problems –** Work quality may become poor due to pressure
iii. **Budget issues –** Costs increase without proper planning
iv. **Lack of honesty –** Not clearly informing stakeholders about changes
v. **Stress on team –** Extra work creates pressure and burnout
vi. **Unfair expectations –** Clients may expect more work without paying more

17. Compare and contrast CPM, and PERT analysis in project scheduling and costing?

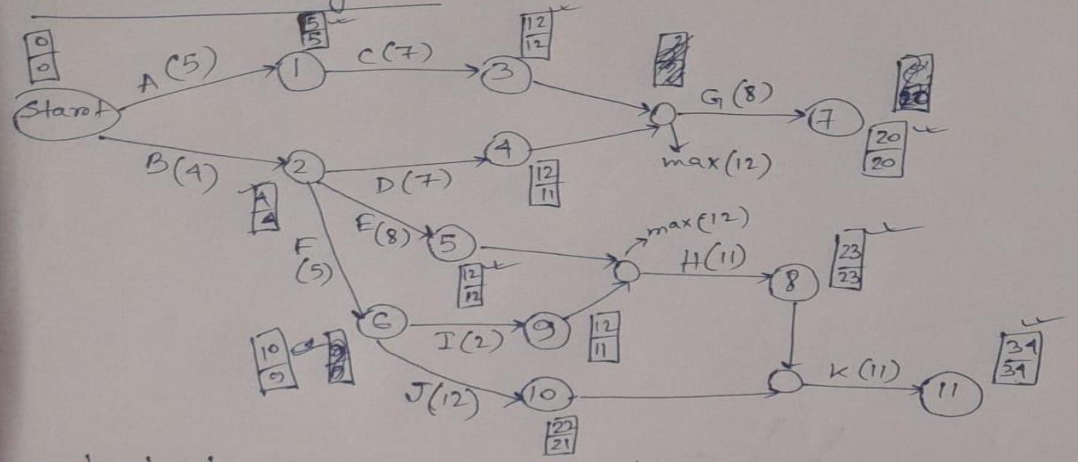
Ans:

CPM	**PERT**
A method used to plan and control projects	A method used to plan and analyse task in a project
Uses exact time for each activity	Uses estimated time for each activity
Nature – Certain	Nature - Uncertain
Simple and easy to use	More complex
Used in construction and manufacturing projects	Used in research, development and non-construction project

18.

Activity	A (5)	B(4)	C(7)	D(7)	E(8)	F(5)	G(8)	H(11)	I (2)	J (12)	K (11)
Pre.	–	–	A	B	B	B	C,D	E,I	F	F	H,J
State	1	2	3	4	5	6	7	8	9	10	11

I. Network Diagram :-



	A	B	C	D	E	F	G	H	I	J	K
LT	5	4	12	12	12	10	20	23	12	22	34
ET	5	4	12	11	12	9	20	23	11	21	34
Slack = (LT − ET)	0	0	0	1	0	1	0	0	1	1	0

Critical Path = Start → 2 → 5 → 8 → 11

⇒ B → E → H → K

& Project Durat^n = 34